

WaterSec AI Agent — Hackathon Challenge Brief

Duration: 24 hours

Format: On-site / Team-based

Domain: Water Intelligence

Context

WaterSec is a water monitoring platform that collects real-time consumption data from IoT sensors installed across various sites and customer profiles. Data flows from physical devices through a processing pipeline and is stored in a structured dataset covering multiple source types, categories, and customer segments.

The goal of this hackathon is to build an **LLM-powered Agent** that allows a user to interact with this water consumption data through natural language — querying it, analysing it, comparing sources, detecting patterns, and generating actionable insights. The agent can respond in any form that best serves the request: a text summary, a data table. **A graph, or a visual comparison (if possible)**

What You Are Given

Dataset (CSV):  Hackathon_dataset

A single pre-processed CSV file containing all consumption records with full metadata. Data Profiles Included

The dataset intentionally covers four distinct real-world customer profiles:

1. Gym — Shower Block: 4 cabins

Each cabin has hot and cold measured through watersec sensor

2. Toilet Block

- Single sensor on the total toilet block (aggregated):
customerA: each deviceID is a toilet BLOC which Covers: 2 flushes, 2 flexible and two sinks
customerB: single deviceID covers: 2 WC (man/woman) each WC has one central sink, 2 flushes, 2 flexible, one وضوء sink

3. Residential CustomerC — Granular dataset

- Most detailed profile: bathroom sink, toilet flush and flexible (sub-category: Tap) each on separate sensors

The Challenge

Build an on-premises conversational AI Agent that can reason over this dataset and respond intelligently to natural language prompts from a user.

What the Agent Can Do — Examples

Teams define the scope and ambition of their agent. The following axes illustrate the expected intelligence:

- **Query & Retrieval** — understand a natural language request and fetch the right data slice from the dataset
- **Analysis & Comparison** — compute, aggregate, and compare across sources, categories, time ranges, or customer profiles
- **Pattern & Anomaly Detection** — surface behaviours, trends, or deviations that are meaningful in a water consumption context
- **Insight & Recommendation** — go beyond the data and produce an actionable, domain-aware output

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Example Prompts the Agent Should Handle

The examples of prompts below are basic examples and the competition result should not be limited to them.

- *"Generate a table comparing shower cabin 1 and cabin 2 between March 10 at 9:00 and April 11 at 10:00"*
- *"Plot the daily consumption trend for the osmosis unit over the past 2 weeks"*
- *"What is the average daily cold water consumption in the gym for the last 30 days?"*
- *"Is there any unusual behaviour on the residential toilet flush this month?"*
- *"Which source has the highest hot water consumption across all sites?"*
- *"Compare bathroom vs kitchen usage for the residential customer in March and visualise it"*

The prompts that requires deeper/advanced level of reasoning and innovation are the ones that will distinguish candidates

- Example: Identify recurrent usage patterns in the Residential Customer dataset. For example, a toilet flush is often followed by bathroom sink usage. Can you detect and describe such patterns in the data?

Technical Expectations

Mandatory

- The LLM must not rely on subscription based AI modes (Claude, GPT, Gemini...)
- Teams are free to choose any model, framework, orchestration layer, or data querying approach they see fit
- A working **system prompt or equivalent mechanism** that gives the model water domain awareness

Evaluation Criteria

Criterion	Weight
Innovative thinking and new datasets used for enhancement	30%
Data accuracy — are the values and comparisons it returns correct?	20%
Agent reasoning and prompt quality — does it understand the intent behind the prompt?	20%
Code quality, architecture and Output quality — clarity, format appropriateness	15%
Water domain intelligence — does it speak the language of water monitoring?	15%

Deliverables

At the end of 24 hours, each team must present:

1. A **running demo** — live agent responding to at prompts from the judges
2. A **brief architecture walkthrough** — presentation: how the agent is structured, what **data** and tools are used